MSCF Probability – Final Exam – August 17, 2017

Name ______________________________

Email ______________________________

Instructions:

- You should complete all questions on this exam by writing out the solutions on paper that will be provided to you. Anything written on the exam paper itself will **not** be graded.
- You will have two hours to complete this exam. No matter when you start the exam, you have to stop at 3:30 PM.
- Notes (except described below), books and calculators/laptops/phones are not allowed.
- A table of common distributions, with pdfs/pmfs, and other information, is included at the end of this exam.
- Final answers can involve any of the following notation:

$$\binom{n}{k},\ n^k,\ \exp(k),\ n!,\ \Gamma(x),\ \Phi(x),$$

  where $\Phi(x)$ is the CDF of the standard normal distribution. Except for cases in which it is explicitly stated, final answers should **not** include integrals.
- You may use 2 handwritten sides of a 8.5" by 11" page (1 double-sided or 2 single-sided) of notes.
- Questions are not all worth the same number of points. The numbers in **bold** after each question indicate the number of points. Also, you may find some easier than others. You should prioritize your work on particular questions accordingly.
  $\boxed{\textbf{Note:}}$ **The total number of points is 48.**
- You **must show all of your work**, unless otherwise stated. A correct answer without explanation will receive only minimal or no partial credit. Note that the "easier" a question may seem, the more burden you have to show that you understand how you are reaching the answer; again, **be clear and complete in your work**.

PLEASE CHECK THAT YOU HAVE PAGES 1 – 6,
THE DISTRIBUTION TABLE, AND THE MATH FACTS

Grading Guidelines:

The following are reasons why you may lose points:

1. Failure to mention when an independence assumption is utilized.
2. Failure to mention when the assumption that random variables are uncorrelated is utilized.
3. Failure to specify clearly the support of a distribution.

You do **not** need to cite the names of rules and definitions when you use them, such as

1. The Rule for the Lazy Statistician
2. The Iterated Expectation Rule
3. The Rule for Conditional Probability
4. The Addition Rule
5. The Chain Rule
6. ...

But, in general, any work that does not make it clear how the answer was found is subject to losing points. In other words, even if you do not justify every step, there should be sufficient number of steps to make it clear you know how to do the derivation. Lack of clarity can also result from messy writing: write neatly.

Part I: For each of the following, simply provide the answer. It is not required that any derivation or justification be provided. Each of these is worth three points.

1. Suppose that A and B are **disjoint** events (i.e., $A \cap B = \varnothing$), and that $P(A) > 0$ and $P(B) > 0$. Which of the following statements **must** be true?

   YOU CAN CHOOSE MORE THAN ONE.

   (a) A and B are dependent (not independent). MUST BE TRUE

   (b) $P(A \text{ or } B) = P(A) + P(B)$ MUST BE TRUE

   (c) $P(A \text{ and } B) = P(A)P(B)$

   (d) $P(A \text{ and } B) < P(A)P(B)$ MUST BE TRUE

   Comment: You either receive full credit or zero points for this question.

2. Suppose that $Z_1, Z_2, \ldots, Z_5$ are i.i.d. random variables, each with the standard normal distribution. Write out an expression for the probability that exactly four of these random variables have value at least 1.5 (and the other is less than 1.5).

$$\binom{5}{4} \Phi(1.5) \left(1 - \Phi(1.5)\right)^4$$

3. Suppose that X has the exponential(λ) distribution. Suppose that a and b are any positive constants. Which of the following are implied by the *memoryless property* of the exponential distribution? **(CHOOSE ONE)**

   (a) $P(X > a + b | X > a) = P(X > a)$

   (b) $P(X > a + b | X > b) = P(X > a) \Big/ P(X > b)$

   (c) $P(X > a + b | X > b) = P(X > a)$ CORRECT

   (d) $P(X > a + b | X > b) = P(X > b)P(X > a)$

4. Suppose that events arrive as a Poisson Process with rate λ. (λ is in units of minute^{-1}.) Let X denote the number of events during two minute period.

   What is the variance of X?

   2λ

Part II: Unless stated otherwise, you need to show your work for these questions, partial credit will be given when appropriate. **Be clear in your steps, and justify them.**

1. Suppose that X has pdf given by

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right), \quad -\infty < x < \infty, \ b > 0.$$

The moment generating function (MGF) for X is

$$M_X(t) = \frac{1}{1 - b^2t^2} \quad \text{for } -1/b < t < 1/b$$

Note: X is said to have the *double exponential distribution* or *Laplace distribution*.

Suppose that Y_1 has the exponential(λ_1) distribution, and Y_2 has the exponential(λ_2) distribution, where λ_1 and λ_2 are positive constants. Assume Y_1 and Y_2 are independent. Define $Z = \lambda_1 Y_1 - \lambda_2 Y_2$. Show that Z has the double exponential distribution with $b = 1$. **(3)**

Get the MGF of the exponential distribution from the table, and then apply properties of the MGF to get the result.

2. Suppose that (X, Y) is uniformly distributed on the unit circle, i.e.

$$f_{(X,Y)}(x, y) = \begin{cases} 1/\pi, & \text{if } x^2 + y^2 \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) What is the marginal distribution of X? **(3)**

Comment: Due to the symmetry of the distribution, X and Y have the same marginal distributions. You do not need to prove this fact.

$f_X(x) = \frac{2}{\pi}\sqrt{1 - x^2}$ for $-1 \leq x \leq 1$ and zero otherwise.

(b) In this case, $E(X) = E(Y) = 0$. (You can use this fact without proof.)

What is the covariance between X and Y? **(3)**

0

(c) Are X and Y independent? Remember to justify your response clearly. **(3)**

They are dependent. You can note the shape of the support, or that the product of the marginals will not equal the joint pdf.

3. There are two urns. Urn 1 contains four red balls and two white balls. Urn 2 contains three red and three white balls. One ball is drawn from Urn 1, hidden from view, and then placed in Urn 2 (so that Urn 2 now has seven balls). Then a ball is drawn from Urn 2. (You can assume that whenever a ball is drawn from an urn, eqch ball in the urn is equally likely to be drawn.)

   (a) What is the probability that the ball drawn from Urn 2 will be red? **(3)**

   11/21

   (b) Given that the ball drawn from Urn 2 is red, what is the probability that the ball transferred from Urn 1 to Urn 2 was red? **(3)**

   8/11

   (c) What is the probability that both balls drawn (the ball from Urn 1 and the ball from Urn 2) are red? **(3)**

   8/21

4. Suppose that (X, Y) has joint pdf given by

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{6}(x + 4y)\,, & \text{if } 0 < x < 2,\ 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

   What is the probability that $X \leq 1$ given that $Y = 0.25$? **(3)**

   3/8

5. Suppose that events of Type I arrive as a Poisson process with rate λ_1, and that events of Type II arrive as a Poisson process with rate λ_2. These two processes are independent: The occurrence of events of Type I has no effect on the occurrence of events of Type II, and vice versa. Also, λ_1 and λ_2 are in the same units.

   Choose some arbitrary time point to serve as the "starting point." Recall that due to the memoryless property of the exponential distribution, this choice is irrelevant.

   What is the expected value of the number of events of Type I prior to the first event of Type II? **(3)**

   Use iterated expectation. Answer is $\lambda_1 \big/ \lambda_2$.

6. Suppose that $X_1, X_2, \ldots, X_{40}$ are i.i.d., each with the Uniform$(0, 1)$ distribution.

   (a) Show that the expected value of $\overline{X} = \sum_{i=1}^{40} X_i/40$ is 0.5. **(2)**
   0.5

   (b) Show that the variance of $\overline{X}$ is $1/480$. **(2)**
   $V(X_i) = 1/12$, so $V(\overline{X}) = 1/480$.

   (c) What is, approximately, $P(\overline{X} \leq 0.4)$? **(3)**
   $\Phi((0.4 - 0.5)/\sqrt{1/480})$.

   (d) Which of the following is true? **(2)**
   Recall that $X_{(1)}$ denotes the minimum of $X_1, X_2, \ldots, X_{40}$.
   Hint: No calculations should be needed.
   CHOOSE ONE:

      i. $P(X_{(1)} \leq 0.4)$ is much smaller than $P(\overline{X} \leq 0.4)$.
      ii. $P(X_{(1)} \leq 0.4)$ is much larger than $P(\overline{X} \leq 0.4)$. CORRECT
      iii. $P(X_{(1)} \leq 0.4)$ and $P(\overline{X} \leq 0.4)$ are approximately equal.

Useful Math Facts

1. **Results Regarding e:** For any number a,

$$e^a = \sum_{i=0}^{\infty} \frac{a^i}{i!} \qquad \text{and} \qquad e^a = \lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n$$

2. **Geometric Series:** If $|r| < 1$ and a is an integer,

$$\sum_{k=a}^{\infty} r^k = \frac{r^a}{1-r}$$

3. **The Binomial Theorem:** For $k = 0, 1, 2, \ldots$,

$$(\lambda_1 + \lambda_2)^k = \sum_{i=0}^{k} \binom{k}{i} \lambda_1^i \lambda_2^{k-i}$$

4. **The Gamma Function:**

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x}\, dx$$

For any t, it holds that $\Gamma(t+1) = t\Gamma(t)$, and for $n = 1, 2, \ldots$, $\Gamma(n) = (n-1)!$. Also, $\Gamma(0.5) = \sqrt{\pi}$.

5. **Definite Integrals:**

$$\int_0^\infty e^{-ax}\, dx = \frac{1}{a}, \quad \int_0^\infty x e^{-ax}\, dx = \frac{1}{a^2}, \quad \int_0^\infty x^m e^{-ax}\, dx = \frac{\Gamma(m+1)}{a^{m+1}}$$

$$\int_0^\infty e^{-ax^2}\, dx = \frac{1}{2}\sqrt{\frac{\pi}{a}}, \quad \int_0^\infty x e^{-ax^2}\, dx = \frac{1}{2a}, \quad \int_0^\infty x^m e^{-ax^2}\, dx = \frac{\Gamma((m+1)/2)}{2a^{(m+1)/2}}$$

$$\int_0^1 t^{x-1}(1-t)^{y-1}\, dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \text{The } \textbf{Beta Function}$$

Name	Density/Mass Function	MGF	Mean	Variance
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$, $-\infty < x < \infty,\ \sigma > 0$	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$	μ	σ^2
Log-Normal	$\frac{1}{x\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(\log(x)-\mu)^2}{2\sigma^2}\right)$, $0 < x < \infty,\ \sigma > 0$	—	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$	$e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}$
Gamma	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$, $0 < x < \infty,\ \alpha,\beta > 0$	$\left(\frac{\beta}{\beta-t}\right)^\alpha, \quad t < \beta$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Exponential	$\beta e^{-\beta x}$, $0 < x < \infty,\ \beta > 0$	$\frac{\beta}{\beta-t}, \quad t < \beta$	$\frac{1}{\beta}$	$\frac{1}{\beta^2}$
Beta	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$, $0 < x < 1,\ \alpha,\beta > 0$	$1 + \sum_{k=1}^{\infty}\left(\prod_{r=0}^{k-1}\frac{\alpha+r}{\alpha+\beta+r}\right)\frac{t^k}{k!}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Uniform (continuous)	$\frac{1}{b-a}$, $a \leq x \leq b$	$\frac{e^{bt}-e^{at}}{t(b-a)}, \quad t \neq 0$ $1, \quad t = 0$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Bernoulli	$p^x(1-p)^{1-x}$, $x = 0, 1,\ 0 \leq p \leq 1$	$pe^t + (1-p)$	p	$p(1-p)$
Binomial	$\binom{n}{x}p^x(1-p)^{n-x}$, $x = 0, 1, \ldots, n,\ 0 \leq p \leq 1$	$[pe^t + (1-p)]^n$	np	$np(1-p)$
Negative Binomial	$\binom{r+x-1}{x}p^r(1-p)^x$, $x = 0, 1, \ldots,\ r \in \{1, 2, \ldots\},\ 0 \leq p \leq 1$	$\left(\frac{p}{1-(1-p)e^t}\right)^r, \quad t < \log(1/(1-p))$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Geometric	$(1-p)^x p$, $x = 0, 1, \ldots .\ 0 \leq p \leq 1$	$\frac{p}{1-(1-p)e^t}, \quad t < \log(1/(1-p))$	$\frac{1-p}{p}$	$\frac{(1-p)}{p^2}$
Poisson	$\frac{e^{-\lambda}\lambda^x}{x!}$, $x = 0, 1, \ldots,\ \lambda > 0$	$e^{\lambda(e^t-1)}$	λ	λ
Hypergeometric	$\frac{\binom{A}{x}\binom{B}{n-x}}{\binom{A+B}{n}}$, $x = 0, 1, \ldots, n$	—	$\frac{nA}{A+B}$	$\frac{nAB(A+B-n)}{(A+B)^2(A+B+1)}$

Table 1: Summary of some common distributions in probability and statistics.